# T3 C_-1: general EFT1 RGEs

Matchete/RGBeta symbolic output in mathematical notation

## C_-1

Additional quartics: 7, additional bilinears: 1, extra Yukawa: none.
*General matching and Wilson coefficients are not computed in this symbolic report.*

## One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

$g_1$

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{43{g_1}^3}{6}$$

$g_2$

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{17{g_2}^3}{6}$$

$g_3$

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

$Y_u$

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

$Y_d$

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

$Y_e$

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} = & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ & + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ & + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ & + \frac{3}{2}\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij} \\ & - \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ & - \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

$m_1^2$

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} = & 6\mathrm{muT3GeneralS1S2}\,(\lambda_{11})^* \\ & + 6\lambda_{11}\,(\mathrm{muT3GeneralS1S2})^* \\ & - \frac{9{g_2}^2 m_1^2}{2} \\ & - \frac{3{g_1}^2 m_1^2}{2} \\ & + 4\lambda_{12}^{(1)} m_2^2 \\ & + 6\lambda_{S_1}^{(1)} m_1^2\end{aligned}$$

$m_2^2$

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} = & 6\mathrm{muT3GeneralS1S2}\,(\lambda_{10})^* \\ & + 6\lambda_{10}\,(\mathrm{muT3GeneralS1S2})^* \\ & - \frac{9{g_2}^2 m_2^2}{2} \\ & - \frac{3{g_1}^2 m_2^2}{2} \\ & + 4\lambda_{12}^{(1)} m_1^2 \\ & + 6\lambda_{S_2}^{(1)} m_2^2\end{aligned}$$

$\lambda_1$

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} = & \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\ & - 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\ & - 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\ & - 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\ & + 8\lambda_7\,(\lambda_7)^* \\ & + 4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\end{aligned}$$

$$
\begin{aligned}
&+ 8\lambda_8 \left(\lambda_8\right)^* \\
&+ 4\lambda_5 \left(\lambda_5\right)^* \\
&+ 4\text{lambdaT3GeneralQ110101I1} \left(\text{lambdaT3GeneralQ110101I1}\right)^* \\
&+ \frac{9{g_2}^4}{4} \\
&+ {g_1}^2 \left(\frac{3{g_2}^2}{2} - 3\lambda_1\right) \\
&- 9{g_2}^2\lambda_1 \\
&+ \frac{3{g_1}^4}{4} \\
&+ 12{\lambda_1}^2 \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_1}^{(A)}}^2}{4} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_2}^{(A)}}^2}{4}
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = &+ 8\lambda_7 \left(\lambda_7\right)^* \\
&+ 8\lambda_9 \left(\lambda_9\right)^* \\
&+ 24\lambda_{11} \left(\lambda_{11}\right)^* \\
&+ \frac{9{g_2}^4}{4} \\
&+ {g_1}^2 \left(\frac{3{g_2}^2}{2} - 3\lambda_{S_1}^{(1)}\right) \\
&- 9{g_2}^2\lambda_{S_1}^{(1)} \\
&+ \frac{3{g_1}^4}{4} \\
&+ 4{\lambda_{12}^{(1)}}^2 \\
&+ \frac{{\lambda_{12}^{(A)}}^2}{4} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_1}^{(A)}}^2}{4} \\
&+ 12{\lambda_{S_1}^{(1)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = &+ 8\lambda_8 \left(\lambda_8\right)^* \\
&+ 8\lambda_9 \left(\lambda_9\right)^* \\
&+ 24\lambda_{10} \left(\lambda_{10}\right)^*
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{9{g_2}^4}{4} \\
&+ {g_1}^2 \left( \frac{3{g_2}^2}{2} - 3\lambda_{S_2}^{(1)} \right) \\
&- 9{g_2}^2 \lambda_{S_2}^{(1)} \\
&+ \frac{3{g_1}^4}{4} \\
&+ 4{\lambda_{12}^{(1)}}^2 \\
&+ \frac{{\lambda_{12}^{(A)}}^2}{4} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_2}^{(A)}}^2}{4} \\
&+ 12{\lambda_{S_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)} (6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& + 24\lambda_7\,(\lambda_7)^* \\
& + 3\lambda_{11}(\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* - (\mathrm{lambdaT3GeneralQ110101I1})^*) \\
& + \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}} (4\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + 3\,(\lambda_{11})^* + 2\,(\mathrm{lambdaT3GeneralQ110101I1})^*) \\
& + \mathrm{lambdaT3GeneralQ110101I1} (2\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* - 3\,(\lambda_{11})^* + 4\,(\mathrm{lambdaT3GeneralQ110101I1})^*) \\
& + 6\lambda_5\,(\lambda_5)^* \\
& + \frac{9{g_2}^4}{4} \\
& - 9{g_2}^2 \lambda_{HS_1}^{(1)} \\
& + \frac{3{g_1}^4}{4} \\
& - 3{g_1}^2 \lambda_{HS_1}^{(1)} \\
& + 4\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)} \\
& + 6\lambda_1 \lambda_{HS_1}^{(1)} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 6\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)} \\
& + \frac{3{\lambda_{HS_1}^{(A)}}^2}{4}
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)} (6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& + 3\lambda_{10}(\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* - (\mathrm{lambdaT3GeneralQ110101I1})^*) \\
& + \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}} (4\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + 3\,(\lambda_{10})^* + 2\,(\mathrm{lambdaT3GeneralQ110101I1})^*)
\end{aligned}
$$

$$
\begin{aligned}
&+ \text{lambdaT3GeneralQ110101I1}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 3\left(\lambda_{10}\right)^* + 4\left(\text{lambdaT3GeneralQ110101I1}\right)^*)\\
&+ 24\lambda_8\left(\lambda_8\right)^*\\
&+ 6\lambda_5\left(\lambda_5\right)^*\\
&+ \frac{9{g_2}^4}{4}\\
&- 9{g_2}^2\lambda^{(1)}_{HS_2}\\
&+ \frac{3{g_1}^4}{4}\\
&- 3{g_1}^2\lambda^{(1)}_{HS_2}\\
&+ 4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}\\
&+ 6\lambda_1\lambda^{(1)}_{HS_2}\\
&+ 4{\lambda^{(1)}_{HS_2}}^2\\
&+ 6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}\\
&+ \frac{3{\lambda^{(A)}_{HS_2}}^2}{4}
\end{aligned}
$$

$\lambda^{(1)}_{12}$

$$
\begin{aligned}
16\pi^2\frac{d\lambda^{(1)}_{12}}{d\ln\mu} = &\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + \left(\text{lambdaT3GeneralQ110101I1}\right)^*)\\
&+ \text{lambdaT3GeneralQ110101I1}(\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + 2\left(\text{lambdaT3GeneralQ110101I1}\right)^*)\\
&+ 24\lambda_9\left(\lambda_9\right)^*\\
&+ 9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)\\
&+ 9\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)\\
&+ 3\lambda_5\left(\lambda_5\right)^*\\
&+ \frac{9{g_2}^4}{4}\\
&- 9{g_2}^2\lambda^{(1)}_{12}\\
&+ \frac{3{g_1}^4}{4}\\
&- 3{g_1}^2\lambda^{(1)}_{12}\\
&+ 4{\lambda^{(1)}_{12}}^2\\
&+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_1}\\
&+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_2}\\
&+ \frac{3{\lambda^{(A)}_{12}}^2}{4}\\
&+ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}
\end{aligned}
$$

$\lambda_5$

$$
\begin{aligned}
16\pi^2\frac{d\lambda_5}{d\ln\mu} = &\lambda_5\left(6\,\text{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) + \frac{\lambda^{(A)}_{12}}{6} + \frac{2\lambda^{(A)}_{HS_1}}{3} + 2\lambda^{(A)}_{HS_2}\right)\\
&- 9{g_2}^2\lambda_5\\
&- 3{g_1}^2\lambda_5
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = &\lambda_{HS_1}^{(A)}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&- 64\lambda_7\,(\lambda_7)^* \\
&+ 4\lambda_{11}(\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + (\mathrm{lambdaT3GeneralQ110101I1})^*) \\
&+ 4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + (\lambda_{11})^*) \\
&+ 4\mathrm{lambdaT3GeneralQ110101I1}((\lambda_{11})^* - 2\,(\mathrm{lambdaT3GeneralQ110101I1})^*) \\
&+ \frac{16}{3}\lambda_5\,(\lambda_5)^* \\
&+ {g_1}^2\,\Big(-6{g_2}^2 - 3\lambda_{HS_1}^{(A)}\Big) \\
&- 9{g_2}^2\lambda_{HS_1}^{(A)} \\
&+ \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = &\lambda_{HS_2}^{(A)}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&- 4\lambda_{10}(\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + (\mathrm{lambdaT3GeneralQ110101I1})^*) \\
&- 4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + (\lambda_{10})^*) \\
&+ 64\lambda_8\,(\lambda_8)^* \\
&+ \mathrm{lambdaT3GeneralQ110101I1}(8\,(\mathrm{lambdaT3GeneralQ110101I1})^* - 4\,(\lambda_{10})^*) \\
&+ 16\lambda_5\,(\lambda_5)^* \\
&+ {g_1}^2\,\Big(6{g_2}^2 - 3\lambda_{HS_2}^{(A)}\Big) \\
&- 9{g_2}^2\lambda_{HS_2}^{(A)} \\
&+ \lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{12}^{(A)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{12}^{(A)}}{d\ln\mu} = & 4\text{lambdaT3GeneralQ110101I1} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* \\ & + 4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\text{lambdaT3GeneralQ110101I1})^* \\ & - 64\lambda_9 (\lambda_9)^* \\ & - 4\lambda_{10}(5 (\lambda_{10})^* + (\lambda_{11})^*) \\ & - 4\lambda_{11}((\lambda_{10})^* + 5 (\lambda_{11})^*) \\ & + \frac{4}{3}\lambda_5 (\lambda_5)^* \\ & + {g_1}^2 \left(-6{g_2}^2 - 3\lambda_{12}^{(A)}\right) \\ & - 9{g_2}^2\lambda_{12}^{(A)} \\ & + 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\ & + 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\ & + 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\ & + \lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}\end{aligned}$$

$\lambda_8$

$$\begin{aligned}16\pi^2 \frac{d\lambda_8}{d\ln\mu} = & \lambda_8(6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)) \\ & + 4\lambda_7 (\lambda_9)^* \\ & - 9{g_2}^2\lambda_8 \\ & - 3{g_1}^2\lambda_8 \\ & + 2\lambda_1\lambda_8 \\ & + 8\lambda_{HS_2}^{(1)}\lambda_8 \\ & + 4\lambda_{HS_2}^{(A)}\lambda_8 \\ & + 2\lambda_8\lambda_{S_2}^{(1)}\end{aligned}$$

$\lambda_7$

$$\begin{aligned}16\pi^2 \frac{d\lambda_7}{d\ln\mu} = & \lambda_7(6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)) \\ & - 9{g_2}^2\lambda_7 \\ & - 3{g_1}^2\lambda_7 \\ & + 2\lambda_1\lambda_7 \\ & + 8\lambda_{HS_1}^{(1)}\lambda_7 \\ & - 4\lambda_{HS_1}^{(A)}\lambda_7 \\ & + 2\lambda_7\lambda_{S_1}^{(1)} \\ & + 4\lambda_8\lambda_9\end{aligned}$$

$\lambda_9$

$$16\pi^2 \frac{d\lambda_9}{d\ln\mu} = 4\lambda_7 (\lambda_8)^*$$

$$
\begin{aligned}
&- 9{g_2}^2\lambda_9 \\
&- 3{g_1}^2\lambda_9 \\
&+ 8\lambda_{12}^{(1)}\lambda_9 \\
&- 4\lambda_{12}^{(A)}\lambda_9 \\
&+ {\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^2 \\
&+ 2\lambda_{S_1}^{(1)}\lambda_9 \\
&+ 2\lambda_9\lambda_{S_2}^{(1)} \\
&+ 5{\lambda_{10}}^2 \\
&+ 2\lambda_{10}\lambda_{11} \\
&+ 5{\lambda_{11}}^2 \\
&+ \text{lambdaT3GeneralQ110101I1}^2
\end{aligned}
$$

$\lambda_{10}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{10}}{d\ln\mu} = {} & 4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*) \\
& - 9{g_2}^2\lambda_{10} \\
& - 3{g_1}^2\lambda_{10} \\
& + \lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11}) \\
& + \lambda_{12}^{(A)}\left(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2}\right) \\
& + \lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\text{lambdaT3GeneralQ110101I1}) \\
& + \lambda_{HS_2}^{(A)}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\text{lambdaT3GeneralQ110101I1}}{2}\right) \\
& + 12\lambda_{10}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{11}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{11}}{d\ln\mu} = {} & 4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*) \\
& - 9{g_2}^2\lambda_{11} \\
& - 3{g_1}^2\lambda_{11} \\
& + \lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11}) \\
& + \lambda_{12}^{(A)}\left(-\frac{\lambda_{10}}{2} - \frac{5\lambda_{11}}{2}\right) \\
& + \lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\text{lambdaT3GeneralQ110101I1}) \\
& + \lambda_{HS_1}^{(A)}\left(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} + \frac{\text{lambdaT3GeneralQ110101I1}}{2}\right) \\
& + 12\lambda_{S_1}^{(1)}\lambda_{11}
\end{aligned}
$$

$\lambda_{12}$

$$
16\pi^2\,\frac{d\lambda_{12}}{d\ln\mu} = \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(6\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))
$$

$$
\begin{aligned}
&+ \lambda_9(8\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 4\left(\text{lambdaT3GeneralQ110101I1}\right)^*)\\
&- 9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\\
&- 3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\\
&+ 2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\\
&+ \lambda_{12}^{(A)}\left(\text{lambdaT3GeneralQ110101I1} - \frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}\right)\\
&+ \lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\text{lambdaT3GeneralQ110101I1})\\
&+ \lambda_{HS_1}^{(1)}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{11})\\
&+ \lambda_{HS_1}^{(A)}\left(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} + \text{lambdaT3GeneralQ110101I1}\right)\\
&+ \lambda_{HS_2}^{(1)}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{10})\\
&+ \lambda_{HS_2}^{(A)}\left(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} - \text{lambdaT3GeneralQ110101I1}\right)
\end{aligned}
$$

### $lambdaT3GeneralQ110101I1$

$$
\begin{aligned}
16\pi^2\,\frac{d\,lambdaT3GeneralQ110101I1}{d\ln\mu} = {}& \text{lambdaT3GeneralQ110101I1}(6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))\\
&+ \lambda_9(8\left(\text{lambdaT3GeneralQ110101I1}\right)^* - 4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)\\
&- 9{g_2}^2\text{lambdaT3GeneralQ110101I1}\\
&- 3{g_1}^2\text{lambdaT3GeneralQ110101I1}\\
&+ 2\lambda_{12}^{(1)}\text{lambdaT3GeneralQ110101I1}\\
&+ \lambda_{12}^{(A)}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\text{lambdaT3GeneralQ110101I1}}{2}\right)\\
&+ \lambda_1(4\text{lambdaT3GeneralQ110101I1} - 2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})\\
&+ \lambda_{HS_1}^{(1)}(4\text{lambdaT3GeneralQ110101I1} - 6\lambda_{11})\\
&+ \lambda_{HS_1}^{(A)}\left(-\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} - 2\text{lambdaT3GeneralQ110101I1}\right)\\
&+ \lambda_{HS_2}^{(1)}(4\text{lambdaT3GeneralQ110101I1} - 6\lambda_{10})\\
&+ \lambda_{HS_2}^{(A)}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} + 2\text{lambdaT3GeneralQ110101I1}\right)
\end{aligned}
$$

### $muT3GeneralS1S2$

$$
\begin{aligned}
16\pi^2\,\frac{d\,muT3GeneralS1S2}{d\ln\mu} = {}& 12\lambda_9\left(\text{muT3GeneralS1S2}\right)^*\\
&- \frac{9{g_2}^2\text{muT3GeneralS1S2}}{2}\\
&- \frac{3{g_1}^2\text{muT3GeneralS1S2}}{2}\\
&+ 2\lambda_{12}^{(1)}\text{muT3GeneralS1S2}\\
&- \frac{3\lambda_{12}^{(A)}\text{muT3GeneralS1S2}}{2}\\
&+ 6\lambda_{10}m_2^2
\end{aligned}
$$

$$+ 6\lambda_{11} m_1^2$$